



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This Report examines the lateral dynamics and gust response of an aircraft and builds on prior work [1], focusing on utilizing Matlab System Identification Toolbox as well as simulating aircraft response to gust inputs. The primary lateral modes analysed in this report is the side-slip (β) and bank angle (ϕ) of the aircraft, and are evaluated through the influence of aileron/rudder control inputs and gust disturbances. This report also evaluates the impact of sensor noise on the the system identification model and methods of improving the robustness of the system identification process. Additionally, the gust/turbulence was modelled using the Dryden Spectra method to aid in simulating the gust responses, in order to compute the maximum side-slip and bank angle perturbations. These finding contribute to modelling and understanding the aircraft's dynamics and control in the lateral motion.

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4.3 Pitch Rate Response

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